The modern parallax:

the redshift-isotropy floor, the empirical forcing of cosmic time and uniform expansion, and the measured resolution of the relativity of simultaneity

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Abstract

The standard cosmological model installs a global cosmic time, a preferred rest frame, and a spatially uniform expansion as kinematical assumptions, prior to and independent of the dynamical fit to data. We show that these assumptions are not conventions the data merely tolerate but structures the data force. The cosmological redshift is the path integral of the expansion rate, $\ln(1+z) = \int H\, dt$, so the observed anisotropy of the cosmic microwave background (CMB) separates exactly into a source term, fixed at last scattering, and a cumulative term, set by the integrated expansion along each line of sight. Granted even perfect homogeneity at last scattering, the observed isotropy of the monopole is then a direct measurement of uniform expansion—so the common supposition that the isotropy is supplied by homogeneity at decoupling is a category error. We make the alternative quantitative: were the expansion rate to track the inhomogeneous matter distribution region by region, with local rate $H(1 - f\delta/3)$ and no single global scale factor, the redshift accumulated along each ray would scatter across the sky by $\sim 10^{-2}$ on the Planck 2018 matter spectrum, and its angular power would exceed the measured temperature spectrum at every multipole from $\ell = 2$ to 2500 by more than four orders of magnitude at $8\,h^{-1}\,\mathrm{Mpc}$ smoothing. Even with the matter taken as uncorrelated from cell to cell along the ray, the most accommodating case, the scatter is 8.9×10^{-3}, some two hundred times the entire measured anisotropy of the CMB. The effect is distinct from the Sachs–Wolfe anisotropies and not a re-derivation of them: when lumpiness clusters upon a single background, a photon's gravitational redshift descending into a potential well is undone climbing out, so the contributions *telescope* along the path and leave no $\sqrt{N}$ accumulation, whereas a genuine differential expansion accumulates because there is nothing to cancel against—so the observed isotropy selects the telescoping picture over the accumulating one. Every choice in the estimate biases it downward, so the number is a floor. A differential programme can evade it only by requiring the local rates to average to a common value in every direction, which is itself a global uniform expansion in a cosmic time—the very structure such programmes set out to dispense with. What is excluded is an inhomogeneous expansion *rate*; the matter lumpiness is untouched, the density being the bend of the spatial cut, so uniform expansion and lumpy matter are consistent. The remaining, finely tuned, observer-centred escape is closed by the Copernican principle together with the independently measured isotropy of the expansion history. What the exclusion selects is a global, time-ordered expansion—the cosmic foliation made dynamical—which is logically prior to the notions of space, isotropy, and homogeneity the model otherwise assumes. The same datum thus establishes, from the bottom up, the cosmic time, the uniformity of its advance, and—within the observable region, under the Copernican principle—the maximal symmetry of its slices. Read ontologically, this measured foliation is the *lapse* of the 3+1 split—an objective rate of advance, the cosmic present—while the relativity of synchrony is the *shift*; the century-old reading of synchrony's relativity as the absence of any objective present is thereby a modal fallacy, which the measured isotropy falsifies outright. We prove that fixing the physical foliation and reading it so is both necessary and sufficient for a coherent formal description of an existing, evolving world—an augmentation of general relativity that changes none of its equations, its necessary half here *measured. We also set down the history the measurement closes.* The structure this datum forces was posited by Einstein in February 1917—against the spirit of relativity, and on

the empirical ground that stellar proper motions are small compared with c—countered a month later by de Sitter, who placed the choice among candidate universes outside physical argument altogether and pressed as his sharpest objection that in Einstein's solution "time has a separate position"; and defended in 1920 by Eddington on geometric grounds, in the declared absence of any experimental knowledge on cosmical scales. *Einstein then went nearly silent on cosmology for the rest of his life*, and never addressed how the cosmic time his own assumption distinguished stands to the relativity of simultaneity. *The assumption was correct*, and the objection was correct as a description of the structure though not as a reason to reject it. This is the empirical counterpart of the conceptual results of the companion papers [4, 6]: as stellar parallax converted the Earth's motion from an interpretive option into a measured fact, the isotropy of the cosmological redshift converts the cosmic foliation from a modelling convenience into a measured feature of the world.

1 Introduction

The standard model of cosmology is assembled in a definite order [24, 25, 26, 27]. One first posits that the average rest frame of matter—made manifest in the redshift–distance relation [22]—defines a coherent cosmic time [42], the one direction in which the four-dimensional world is, in Eddington's phrase, not curved [23]; then that space is the family of hypersurfaces orthogonal to it; then that this space is isotropic; then that it is homogeneous; and only last that its geometry evolves by a scale factor $a(t)$ whose form the field equations fix through the energy content [29, 30]. The first four are kinematical posits; the dynamical fit comes fifth and last. The standard presentation treats the kinematical posits as idealizations vindicated after the fact by the success of the fit—a global cosmic time, a preferred rest frame, and a uniform expansion accepted because the model built on them works. The nearest argument in the literature is of a different kind. The Ehlers–Geren–Sachs theorem and its almost-isotropic extension deduce the geometry from a symmetry: if every comoving observer sees an (almost) isotropic radiation background, the spacetime is (almost) Friedmann [43, 44]. Its premise is isotropy at every observer, which cannot be observed from here, and the almost-isotropic form needs bounds on the derivatives of the anisotropies as well, without which an almost isotropic temperature does not imply an almost isotropic universe [45]. Neither reads the isotropy we do observe as what it also is: the record of the radiation's journey to us.

The claim of this paper is that observation forces the cosmic time and the uniformity of the expansion it carries and, with the Copernican principle, the isotropy and homogeneity of its slices, and that it forces them directly rather than through the success of the dynamical fit. It does so by falsifying the alternative. (The second posit, that the slices are orthogonal to the matter's worldlines, is not fixed by the measurement alone; §5 records that qualification.)

The observable is the isotropy of the CMB, and the argument turns on what that isotropy is made of. The temperature received from each direction is the temperature at emission divided by the redshift accumulated on the way, so the observed anisotropy is the sum of two terms (§2): a *source* term, the Sachs–Wolfe and acoustic anisotropies fixed at last scattering, and a *cumulative* term, the directional variation of the expansion the radiation crossed in transit. The two are independent, and the observed isotropy constrains both. The standard reading attributes the anisotropy to the source term alone, the cumulative term having been set to zero in advance by the posit of a single scale factor, so it is never put to the test. Read as a sum, the isotropy is also a measurement of the journey: the radiation reaches us across the whole of the expansion since decoupling, redshifted by a factor of 1089 in every direction, and it carries the 10^{-5} primordial pattern to us intact.

The alternative to a uniform expansion is that the expansion tracks the matter: that regions of different density expand at different rates, as the averaging programmes of inhomogeneous cosmology hold. If it does, the redshift a photon accumulates depends on the regions it crosses, and directions that cross different structure arrive with different redshifts. This contribution is

distinct from the Sachs–Wolfe effects, and the distinction is the crux (§3.5). A potential well on a single background shifts a photon's energy on the way in and returns the shift on the way out, so the gravitational contributions *telescope* along the path and leave only endpoint and integrated residues; a differential expansion has nothing to cancel against, and its contribution accumulates. What is at issue is the expansion *rate*, not the matter: the matter can be as lumpy as it is observed to be, provided it clusters upon a single expansion (§3.6).

We isolate the cumulative term and compute it under that hypothesis, tying the local expansion rate to the observed matter distribution through the Planck 2018 matter spectrum (§3). Each simplification in the estimate lowers it: the full matter coefficient retained, the contrast weighted by its linear growth, linear theory for the matter and for the local rate, statistically independent cells along the ray in its most accommodating form, and a comparison that attributes the whole of the observed anisotropy to this one term, as though the primordial and Sachs–Wolfe contributions were absent. The result is therefore a floor. It exceeds the measured anisotropy by more than two orders of magnitude in amplitude and four in angular power, so the premise that the expansion has tracked the matter is false. Where the standard argument runs only the affirmative half of the scientific loop—the model is consistent with uniform expansion—this runs the half that decides: the alternative is given its observable consequence, and the data reject it.

The argument is made from what we observe, along our own past light cone. The Copernican principle enters only to close the one remaining escape, an inhomogeneity finely tuned about our position, and to carry the result from our light cone across each slice (§4).

This is the modern counterpart of a familiar episode. Before Bessel, whether the Earth moved was, on a strict reading, an interpretive choice among empirically equivalent pictures; stellar parallax converted it into a measured fact [88], and the heliocentric frame from an option into the one the measurement singled out [12]. A purely relational reading of relativity holds, by the same token, that a preferred cosmic rest frame could never be observable even if it existed. The CMB is the measurement that overturns that expectation: the universe furnished, through the physics that produced the background and the geometry of its uniform expansion, a direct measurement of our motion with respect to the cosmic rest frame, and with it of the cosmic time that frame carries [10, 11]. The dipole was predicted as the signature of that motion [46], detected [47] and is now measured to a fraction of a per cent [48], and the aberration and Doppler boosting it imprints on the small-scale anisotropies confirm it as a motion [49]. Whether the mean rest frame of the matter coincides with that of the CMB is tested independently by the dipole in source counts [50], where an amplitude exceeding the kinematic expectation has been reported [51, 52]; that test bears on how the matter's rest frame stands to the cosmic foliation (§5), not on the uniformity of the expansion measured here. What follows makes that "modern parallax" quantitative.

The result bears on three lines of work in which that foliation has been treated as optional. In inhomogeneous cosmology, the averaging, or backreaction, programme [19, 53, 54, 55, 56, 57] lets regions of different density expand at different rates on a common foliation, and the floor excludes that premise. That includes the timescape model [20, 38], which makes the expansion uniform in each region's own clock but not along the journey of the light (§3.5); an inhomogeneous cosmology that survives keeps a uniform expansion. In the 3+1 formulation of general relativity [70] the lapse and shift are freely specifiable, slicings are chosen for computational convenience [71], and the absence of a preferred foliation underlies the problem of time in canonical quantum gravity [72, 73]; the data single one out, and where the matter's worldlines are orthogonal to it, it is the foliation of constant mean curvature [68] (§5). And in the philosophy of time, the Rietdijk–Putnam argument [77, 78] and Penrose's Andromeda paradox [80] infer from the relativity of simultaneity that no physically distinguished present exists; in our universe one is distinguished, and measured (§5).

A real, global cosmic time is a structure general relativity's formalism admits but does not

supply, and a companion paper takes it as a commitment on other grounds [4]. This paper shows that the commitment is not merely admissible but empirically forced. *The two arguments are deliberately independent*: a companion result on the causal occurrence of gravitational collapse stands on causal structure alone and needs none of the cosmological evidence below [6], and nothing in the present argument draws on it. That they meet is therefore evidence and not construction. We stress the register at the outset, as the companion papers do: nothing here modifies general relativity. The equations are unchanged. What is established is which of the formally available foliations is the physical one—and, read ontologically (§5, Theorem 1), that this physical foliation is the *lapse*, the objective cosmic present, while the relativity of synchrony is the *shift*: the measured isotropy thereby settles the century-old question the relativity of simultaneity left open—synchrony relative, simultaneity absolute—exposing the reading that denied an objective present as a modal fallacy, and fixing this reading as the necessary and sufficient augmentation of general relativity for a coherent description of an existing, evolving world.

2 The two contributions to the observed anisotropy

The redshift along a line of sight is the total expansion the photon underwent, $1+z = a_{\rm obs}/a_{\rm em}$, equivalently $\ln(1+z) = \int H\,dt$ integrated along the ray: the cosmological redshift simply *is* the accumulated expansion in that direction, a path integral and not a property of the source. *That path integral is a conserved quantity written out, which is why it is a property of the path.* In conformal form $ds^2 = a^2(\eta)(-d\eta^2 + d\mathbf{x}^2)$ the vector ∂_η is a *conformal* Killing vector, so its charge $\xi\cdot p$ is conserved along a geodesic only where $p\cdot p = 0$; on the null cone that charge is $a\omega$, and its constancy *is* $1+z = a_{\rm obs}/a_{\rm em}$, whose logarithm is $\int H\,dt$. *The restriction to the null cone is not incidental to the argument below but the reason it works*: for a massive carrier $p\cdot p = -m^2$, the charge is not conserved, and no such line-of-sight integral exists—so the redshift is a statement about light and could not have been obtained from any other messenger[L14]. The observed temperature is $T_{\rm obs}(\hat n) = T_{\rm em}(\hat n)/(1+z(\hat n))$, so the observed anisotropy separates exactly into two physically distinct contributions,

$$\frac{\delta T_{\rm obs}}{T_{\rm obs}} = \frac{\delta T_{\rm em}}{T_{\rm em}} \;-\; \delta\!\int H\,dt, \tag{1}$$

a *source* term fixed by conditions on the last-scattering surface, and a *cumulative* term set by the directional variation of the integrated expansion rate over the whole path. The two are independent: the value of one constrains the other not at all. Observed isotropy therefore requires *both* to be small—uniformity at decoupling *and* uniformity of the subsequent expansion—and these are precisely the two kinematical assumptions the standard model installs by hand, homogeneity at decoupling zeroing the first term and a single direction-independent scale factor $a(t)$ zeroing the second.

The decisive consequence is seen by granting the common reading its strongest form. Suppose the last-scattering surface were perfectly homogeneous, $\delta T_{\rm em} = 0$ exactly. Then (1) reads $\delta T_{\rm obs}/T_{\rm obs} = -\,\delta\!\int H\,dt$, with nothing left over: the entire observed isotropy is the statement that the integrated expansion was the same in every direction. Homogeneity at the source does not *supply* the isotropy; granted perfectly, it removes one term and leaves the observed isotropy as a *direct measurement of uniform expansion.* The surface is not in fact homogeneous, and that sharpens the point: its Sachs–Wolfe and acoustic anisotropies, at the 10^{-5} level, arrive with their predicted spectrum intact. The journey that carried them imposed a redshift of 1089, and it imposed it the same in every direction to well within one part in 10^5; otherwise it would have overwritten the primordial pattern. The widespread supposition that the monopole's isotropy is delivered by homogeneity at last scattering, full stop, is thus a category error—it credits the source alone for a fact about the *observed* radiation that resides in both terms, the second of

which only the intervening expansion can supply. What the remainder of this paper does is put a number to the cumulative term under the alternative to a global $a(t)$, an expansion that tracks the matter, and read that number against the data.

3 The redshift-isotropy floor

3.1 The hypothesis under test

The hypothesis the floor tests is that the large-scale expansion rate tracks the matter lumpiness region by region—that the expansion follows the local density, with no single global scale factor. This is the premise of the averaging, or backreaction, programme, in which domains of different density expand at different rates on a common foliation [19, 53, 54, 55, 56, 57], and of treatments of light propagation in which the redshift accumulates the local expansion rate along the ray [18]. The hypothesis is the complement of the case the isotropy of the CMB has long been used to bound: a global anisotropy of the expansion, a shear of the Bianchi type [81, 82]; direction-dependent expansion rates are now also mapped directly, from the CMB and from low-redshift distances [83]. The floor bounds the other way the expansion could fail to be uniform: a rate that varies from region to region with the density. We state it precisely through the general relation for the redshift accumulated along a ray. For a matter congruence with four-velocity u^a, expansion Θ, shear σ_{ab} and acceleration A_a, the redshift measured by its members along a ray with spatial direction e^a accumulates as [16, 17]

$$\frac{d\ln(1+z)}{d\ell} = \tfrac{1}{3}\,\Theta + \sigma_{ab}e^a e^b + A_a e^a, \tag{2}$$

per unit proper distance ℓ along the ray (units with $c = 1$). In a Friedmann model only the first term survives and $\Theta = 3H$. In the hypothesis under test each region expands as its own Friedmann patch at a rate set by its own density—the separate-universe correspondence, in which a spherical region of uniform overdensity evolves as a closed Friedmann patch—so its local expansion is isotropic, its shear vanishes, and for pressureless matter its acceleration vanishes: the redshift accumulates the local expansion rate along the path, $d\ln(1+z) = H_{\rm loc}\,dt$.

To linear order the separate-universe correspondence assigns a region of fractional matter contrast $\delta \equiv \delta\rho/\rho$ a fractional deficit in its local scale factor. Matter conservation gives $\rho \propto a^{-3}$, hence $d\rho/\rho = -3\,da/a$, so

$$\frac{\delta a}{a} = -\frac{1}{3}\,\delta. \tag{3}$$

An overdense region has expanded less; an underdense region more. The local rate is the time derivative of this deficit. With the contrast growing as $\dot\delta = fH\delta$, where $f \equiv d\ln D/d\ln a$ is the growth rate of the linear growth factor D, a region of smoothed contrast δ_R expands at

$$H_{\rm loc} = H\left(1 - \tfrac{1}{3} f\,\delta_R\right). \tag{4}$$

3.2 Accumulation along the line of sight

The observed CMB temperature in a given direction is $T_{\rm obs} = T_{\rm lss}/(1+z)$, so to the order kept $|\delta T/T| = |\delta\ln(1+z)|$: the temperature contrast between directions is the contrast in accumulated expansion. A photon crossing comoving distance $d\chi$ spends $dt = a\,d\chi$ in transit, so the direction-dependent part of its accumulated redshift is

$$\delta\ln(1+z)(\hat n) = -\int_0^{\chi_*} d\chi\; g(\chi)\,\delta_R\big(\chi\hat n,\, z(\chi)\big), \qquad g(\chi) = \tfrac{1}{3}\,aHf, \tag{5}$$

integrated to the last-scattering distance χ_*, the factor $1/3$ from (3) carried in g. Two statistics of this field follow directly from the linear matter power spectrum smoothed by the top-hat

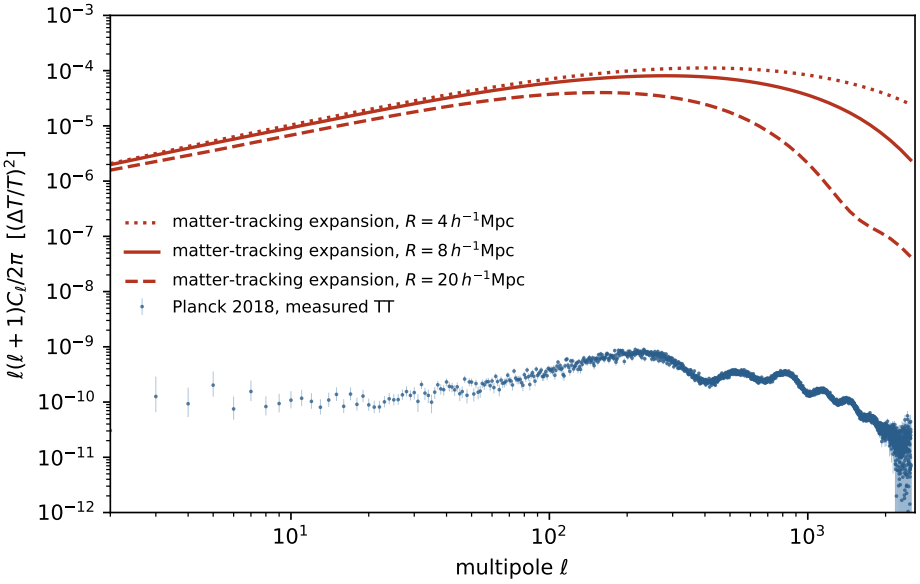


Figure 1: Angular power of the redshift anisotropy that an expansion tracking the matter distribution would produce, from (7) on the Planck 2018 linear matter spectrum, for three smoothing radii (red), against the temperature power spectrum Planck measures (blue) [14]. Both are in units of $(\Delta T/T)^2$. The prediction exceeds the measurement at every multipole, by at least 2×10^4 for $R = 4$ and $8\,h^{-1}$ Mpc, and by at least 800 for $R = 20\,h^{-1}$ Mpc, where the smoothing begins to remove the smallest angular scales.

window W of radius R, $P_R(k,z) = P(k,z)\,W^2(kR)$. The variance along a single ray is

$$\sigma^2_{\rm path} = \int_0^{\chi_*} d\chi\; g^2(\chi)\,\frac{1}{2\pi}\int_0^\infty dk\,k\,P_R(k,z), \tag{6}$$

and its angular power spectrum is, in the Limber approximation [69],

$$C_\ell = \int_0^{\chi_*} d\chi\;\frac{g^2(\chi)}{\chi^2}\,P_R\!\left(\frac{\ell+1/2}{\chi},\,z(\chi)\right), \tag{7}$$

which is directly comparable, multipole by multipole, with the measured temperature spectrum in units of $(\Delta T/T)^2$.

3.3 The number

We evaluate (6) and (7) on the linear matter spectrum of the Planck 2018 base-ΛCDM model [13], computed with CAMB [15] ($\sigma_8 = 0.811$, $z_* = 1089.9$), with the growth rate f taken from the same spectrum. For a smoothing radius $R = 8\,h^{-1}$ Mpc,

$$\sigma_{\rm path} \simeq 1.6 \times 10^{-2}. \tag{8}$$

The rms of the measured CMB temperature anisotropy, summed over $\ell = 2$–2508 from the Planck 2018 spectrum [14], is 4.1×10^{-5}. The hypothesis predicts a scatter in the accumulated redshift alone some 390 times larger than the entire measured anisotropy[P4R1]. Figure 1 makes the comparison scale by scale: the angular power (7) exceeds the measured power at every multipole from $\ell = 2$ to 2500, by a factor of at least 2×10^4 and of 2×10^5 at the median multipole[P4R2].

The choice of R does not rescue the hypothesis. The radius is the scale on which the rate is taken to follow the density, and the rms falls slowly as it grows: 1.9×10^{-2} at $R = 4$, 1.1×10^{-2} at 20, 6.0×10^{-3} at 50, 3.3×10^{-3} at 100 and 9.6×10^{-4} at $300\,h^{-1}\,\mathrm{Mpc}$—still 24 times the whole measured anisotropy at a smoothing scale far above any on which the matter is inhomogeneous[P4R1]. At large smoothing the exclusion moves to the low multipoles: at $R = 50$ and $100\,h^{-1}\,\mathrm{Mpc}$ the predicted power still exceeds the measured power at every multipole to $\ell = 2500$, and at $R = 300\,h^{-1}\,\mathrm{Mpc}$ it exceeds it at every multipole up to $\ell \approx 490$, by more than a factor of 60 up to $\ell = 100$[P4R2].

3.4 Why the number is a floor

Each approximation in the estimate lowers it:

- *The coefficient.* The full matter coefficient 1/3 is retained; dropping it would treble the estimate.
- *Growth.* The contrast is weighted by its linear growth along the path, which suppresses it at high redshift, where most of the path lies; holding σ_R at its present value along the whole path would give an estimate nearly ten times larger.
- *Linear theory.* The matter contrast on the scales that dominate (6) is nonlinear at low redshift, where nonlinear growth raises the small-scale power above the linear spectrum used here; the void and wall contrasts the backreaction programmes invoke are of order unity.
- *The local rate.* Equation (4) is the linear limit of the separate-universe rate; the exact rate of a nonlinear underdensity departs from the mean by more, not less.
- *The comparison.* The measured anisotropy includes the primary acoustic signal, which the hypothesis does not remove; the redshift scatter it predicts must fit inside the measured power together with everything else on the sky, so comparing it with the whole measured spectrum understates the exclusion.

The value (8) carries the correlations of the matter field along the ray, which the measured spectrum shows and structure formation in general relativity requires. The most accommodating value drops them: light crossing statistically independent cells of uniform density, each of comoving length R and with its own contrast $\sigma_R(z)$, as many of them as the finite distance to last scattering holds—$N = \chi_*/R \approx 1200$ at $R = 8\,h^{-1}\,\mathrm{Mpc}$—so that the scatter averages down as far as the observable universe allows. Each cell adds $gR\,\delta_R$ to (5), independent contributions add in variance, and

$$\sigma^2_{\mathrm{cells}} = \int_0^{\chi_*} d\chi\; g^2(\chi)\, R\, \sigma_R^2(z) \tag{9}$$

gives $\sigma_{\mathrm{cells}} \simeq 8.9 \times 10^{-3}$, still some 220 times the whole measured anisotropy[P4R1]. That value is unrealistically uniform, and the rise from it to (8) is a theorem rather than a caution: for N cells of average pairwise correlation $\bar\rho$ the variance of their sum is $N\sigma^2(1 + (N-1)\bar\rho)$, which exceeds the independent value for every $\bar\rho > 0$ and falls below it only for $\bar\rho < 0$[P4R5L24]. Gravitational clustering produces positive correlations along the line of sight, so the expectation general relativity itself supplies for the matter field moves the prediction further from the measured sky, not toward it.

The rms comparison sets an anisotropy, the scatter across directions, against (6) and (9), which are variances along one ray. The two differ only for modes longer than the path: every sightline is cast from the same observer, so a mode much longer than the observable region takes one value over the whole of it, lands on every direction alike, and is removed with the monopole. Measured, the across-sky scatter tracks the single-ray variance to 1% for modes much shorter

than the path and falls to 2×10^{-4} of it for modes much longer[L12]; on a CDM spectrum the fraction of the single-ray variance lying in modes longer than the path is 3×10^{-6}, and 95% of it lies in modes at least seventy times shorter, so the single-ray variance is the across-sky scatter everywhere the variance actually lives[P4R6L24]. The angular power (7) makes the same comparison directly, direction against direction. The Limber approximation is inaccurate at the lowest multipoles, by tens of per cent at $\ell \lesssim 10$, which is immaterial against margins of four orders of magnitude; (6) rests on the same narrow-correlation approximation along the ray, which holds because the correlation length of the smoothed field is short against the distances over which g and P change.

The comparison is a lower bound against an upper limit, which is what makes it robust rather than close: the measured anisotropy bounds from above whatever redshift scatter the sky carries, and every free choice in the estimate pushes the two further apart, so the exclusion cannot be weakened by a different estimate, only by a different physical claim[L25]. For even the most accommodating value merely to reach the measured anisotropy, the sky would have to be some 220 times more anisotropic than it is. The hypothesis that the expansion rate is locally set by the matter distribution is therefore excluded, and not marginally.

3.5 Distinct from the Sachs–Wolfe effect

It is essential that this is a distinct effect from the ordinary Sachs–Wolfe and integrated Sachs–Wolfe anisotropies [31, 60], not a re-derivation of them. In the uniform-expansion picture the matter lumpiness is clustering upon a single background $a(t)$, carried by a metric potential Φ. In (2) the local expansion of the matter still varies, but the shear term no longer vanishes, and to linear order the two combine into a total derivative along the ray: a photon's gravitational redshift descending into a potential well is undone climbing out, so the contributions *telescope* along the path, leaving the endpoint potential ($\sim 10^{-5}$) and the small integrated term, with no $\sqrt{N}$ accumulation. The floor computed here is the contrary case—a genuine differential *expansion*, a cosmological redshift that accumulates because there is nothing to cancel against. The observed anisotropy thus selects the telescoping, uniform-expansion picture over the accumulating, differential one. The isotropy of the monopole is not accounted for by large-scale statistical homogeneity [61]—that homogeneity is fully consistent with the differential-expansion hypothesis, which would still produce the scatter of (8)—but by uniform expansion in a cosmic time.

It is the scatter, not the mean, that is at issue. In statistically homogeneous dust spacetimes the redshift along a long ray tracks the volume-averaged expansion, with the local departure of the rate from its average entering as a fluctuation that is argued to be small when integrated over long distances [18]. The floor is that fluctuation computed: under a matter-tracking rate it is of order 10^{-2} along each ray, and it is this direction-to-direction scatter, not the mean redshift, that the measured isotropy excludes.

A differential-expansion programme can evade the floor only by requiring the local rates to average to a common value in every direction: a uniform large-scale Hubble flow. But that requirement is itself a global uniform expansion in a cosmic time, the very structure such programmes set out to dispense with; imposing it to recover the observed isotropy reinstates the cosmic rest frame and the objective simultaneity it carries. Either way, the observed isotropy of the monopole redshift forces a uniform cosmic expansion.

The timescape model [20, 38, 39] is the most developed realization of the averaging programme, and it falls inside the hypothesis. It divides the universe into voids and walls whose expansion rates, referred to the volume-average time, differ: in its tracker solution the walls expand at two-thirds the rate of the voids, and the difference is the backreaction that produces its apparent acceleration, $\mathcal{Q} = 6 f_{\rm v}(1 - f_{\rm v})(H_{\rm v} - H_{\rm w})^2$, with $f_{\rm v}$ the void fraction. Its further postulate is that each region measures the same expansion rate in its own proper time, the difference being carried by the rates of clocks, which differ between walls and voids by 38 per

cent today. That postulate does not remove the differential expansion from the redshift. By (2) a photon crossing a region gains the local rate per unit of local proper time, and per unit of volume-average time it spends less local proper time crossing a wall than a void; the redshift it gains per unit of volume-average time is therefore the region's own rate, $H_{\rm v}$ or $H_{\rm w}$, while the clock rates enter only at the ends of the path, through the acceleration term. The mean of that accumulation is exactly the model's own redshift relation [39] (its eq. 38), and its scatter is of the floor's kind: for void and wall cells drawn independently along each ray, the most accommodating case, the rms of $\ln(1+z)$ across directions is 2.4×10^{-2}, 4.2×10^{-2} and 5.4×10^{-2} for cells of 10, 30 and $50\,h^{-1}$ Mpc, some 600 to 1300 times the whole measured anisotropy[P4R3]. The timescape papers compute observables on the average geometry [39, 40, 41] and do not compute this scatter. Uniformity in local proper time is not uniformity along the journey of the light, and it is the latter that the isotropy of the CMB measures.

3.6 Uniform expansion, lumpy matter

The floor constrains the expansion *rate*, not the matter content. CR's real space is locally curved and inhomogeneous—the matter density is the bend of the spatial cut off the de Sitter vacuum profile [8, 9]—and that lumpiness is untouched by the argument. What is excluded is an inhomogeneous expansion rate *tracking* the lumpiness: the layer advances uniformly in cosmic time while its intrinsic geometry carries the bend. Uniform expansion and lumpy matter are consistent, and only the former is at issue here.

4 What the exclusion establishes

The floor closes the statistical escape, and one further escape is logically available and must be closed separately. A finely tuned, spherically symmetric inhomogeneity centred on the observer would produce isotropic redshift about us with no global uniform expansion; being structured rather than statistical, it is not touched by the $\sqrt{N}$ floor. It is closed instead by the Copernican principle together with the independent isotropy of the *expansion history* itself: standard rulers (baryon acoustic oscillations [35]) and standard candles (Type Ia supernovae [36, 37]) reconstruct the expansion along many directions and agree across the sky, so the uniformity is not the vantage artefact of a privileged centre [33, 34, 62]. Gpc-scale voids centred on us, the form this escape takes in the literature, are excluded directly by the kinetic Sunyaev–Zel'dovich effect and by the combined distance data [63, 64]. With the statistical alternative excluded by the floor and the centred one by the Copernican step, the disjunction is exhausted: the observed isotropy forces uniform cosmic expansion.

What that exclusion reaches is the point of the paper. The region the data reject is not a parameter corner but the entire class of histories lacking a single global scale factor on a single global time; to reject that class is to select a global, time-ordered expansion, which is the cosmic foliation made dynamical. And the foliation is logically prior to everything the standard model otherwise assumes—"space" is one of its slices, "expansion" an ordered family of them, "isotropy" and "homogeneity" properties of a slice—none of which is even statable until the foliation is in hand. So a single datum establishes, from the bottom up, the cosmic time on which the rest stands, the uniformity of its advance, and—within the observable region and under the Copernican principle—the maximal symmetry of its slices. The model's foundational kinematics are not idealizations the data tolerate; they are what the data force, the one coherent alternative excluded.

5 The history the measurement closes

The structure this datum forces was posited, attacked, defended, and abandoned within four years, and the record is worth setting down because the abandonment shaped a century of practice.

Einstein's cosmological paper of February 1917 assumed, as what he called an "approximative assumption", that *"there is a system of reference relatively to which matter may be looked upon as being permanently at rest"* [28]—an assumption against the spirit of relativity [3]. *His ground was empirical*: "the most important fact that we draw from experience as to the distribution of matter is that the relative velocities of the stars are very small as compared with the velocity of light". In a special-relativistic world one would expect neighbouring bodies at velocities drawn uniformly from $(0, c)$; the observed smallness of proper motions is evidence that the world lines are nearly parallel, and the three-space in which they are at rest is distinguished. *That inference is the one this paper's floor now closes by more than two orders of magnitude*, and it was drawn from stellar velocities a century before.

De Sitter's reply came one month later and countered every consideration in turn [1]. The boundary conditions were dismissed as unobservable in principle and, given the new λ-term, achievable anyway; the world-matter Einstein thought essential was declared to "serve no other purpose than to enable us to suppose it not to exist"; and the choice among the candidate universes was placed outside physics altogether—it "cannot be decided by physical arguments, but must depend on metaphysical or philosophical considerations, in which of course also personal judgment or predilections will have the same influence". *And the criticism he pressed most was the one that matters here*: that in Einstein's solution "time has a separate position" from space.

Three years later the same structure was defended on geometric grounds, by the relativist best placed to do it. Eddington held that "the world taken as a whole has one direction in which it is not curved; that direction gives a kind of absolute time distinct from space", that relativity "is not concerned to deny the possibility of an absolute time, but to deny that it is concerned in any experimental knowledge yet found", and that "it need not perturb us if the conception of absolute time turns up in a new form in a theory of phenomena on a cosmical scale, *as to which no experimental knowledge is yet available*" [23]. The time in question is that of "a being coextensive with the world", which is the foliation. *He names the absence of the discriminator, and selects anyway.*

What followed was silence. The 1917 paper was the last in which Einstein published the details of his cosmological considerations. After Hubble's result he wrote some five pages on the problem in two short articles, the second co-authored with de Sitter, while de Sitter published volumes; and his later rejection of the cosmological term shaped the expansion paradigm for the rest of the century [3]. *He never returned to the question his own assumption raised*—how a cosmic time distinguished by the matter's rest frame stands to the relativity of simultaneity—and so far as the record shows never addressed it in print.

The measurement reported above closes that arc. What Einstein posited as an approximation on stellar velocities, what de Sitter placed beyond physical argument, and what Eddington defended without evidence, is what the redshift isotropy now forces. *The assumption that ran against the spirit of relativity was correct*, and the objection that it gave time "a separate position" was correct too—as a description of the structure, though not as a reason to reject it. *We record the history because the reasoning that abandoned this structure is still in the field's hands*, and because a selection made in 1920 in the declared absence of evidence, met by a measurement in the decades after, is the pattern by which theory-choice earns its standing [7]. *And the parallax of this paper's title should be read with the same care the historical case requires.* Stellar parallax went undetected for two millennia and was read as telling against the Earth's motion, but before the cosmos was given depth the relevant parallax was not a conceivable measurement at all—the stars sat on a sphere and had no distances to be displaced against [7]. The objection is post-Digges, and it is Tycho who presses it. *The lesson is the one this paper's*

floor turns on: an absent discriminator is not an absent fact, and whether a discriminator is even available is a question about the picture in hand rather than about the world.

What the standard model installs as assumptions, the present framework derives. The cosmic-time variable was put in by hand at the outset—assumed in Einstein's 1917 cosmology [28] and carried through Friedmann's expanding solutions [29] to the present model, where a single global scale factor on a single global time is the background fixed before any matter is placed upon it. That foundational structure is here forced by measurement. *A second assumption entered in the same 1917 paper, and it runs in the opposite direction from what the present result establishes.* Einstein's static solution fixes the cosmic scale from the content: with $\lambda = \kappa\rho/2$ the radius is $R = 1/\sqrt{\lambda}$, so *the density of the world-matter determines the size of the world* [28]—and Friedmann's equations carry that direction of determination forward, the scale factor's evolution being sourced by what the universe contains [29]. What the isotropy forces is the reverse ordering for the *rate*: it is fixed independently of how the content is distributed, since a rate tracking the content is precisely what the measurement excludes. Whether the same reversal extends from the rate to the scale is not settled here and is taken up in a companion [4]. This is the empirical motivation the title names: not a modification of general relativity, whose equations are untouched, but the measured selection of the foliation the theory admits and does not otherwise single out.

What the measurement forces is the empirical half of a structural result, and the half in hand fixes the other, so it is worth stating whole. General relativity supplies a four-geometry and field equations but does not single out which of the foliations its formalism admits is the physical one—as the companion papers stress and as the present forcing makes vivid [4]. In the 3+1 form of the theory [70] a slicing is described by two functions: the lapse N, the proper time elapsed along the normals between neighbouring slices, and the shift N^i, the motion of the spatial coordinates from one slice to the next relative to those normals. In the canonical theory both are Lagrange multipliers, freely specifiable. Numerical relativity chooses them for stability, with maximal or 1+log slicing [71], and the absence of any preferred foliation is the root of the problem of time in canonical quantum gravity [72, 73]. Programmes that do posit a preferred foliation posit it on theoretical grounds: the constant-mean-curvature slicing that defines York's extrinsic time [68], which shape dynamics takes as fundamental [74]; the preferred foliation of Hořava–Lifshitz gravity [75]; and the dynamical preferred frame of Einstein–aether theory [76]. The standard interpretation fills the theory's silence on which slicing is physical by collapsing two distinctions at once: it reads the relativity of simultaneity as the absence of any objective present, and it reifies the four-manifold as a thing that exists. Each is a category error of the same family as the one corrected in §2—there, crediting the source alone for a fact that resides also in the intervening expansion; here, crediting the manifold for what the evolving world does. Pried apart, the two are the parts a coherent formal description of an existing, evolving world requires; and they are exactly the lapse and the shift of the 3+1 split read ontologically—the Newtonian part, an absolute rate of advance, and the Einsteinian part, the relativity of synchrony, the formalism the standard reading had fused and a coherent description must hold apart.

The relativity of *synchrony* is general—a feature of relative motion, the light postulate fixing only its magnitude [11, 85]—and it is the shift, the freedom of a moving observer to tilt the slices off the comoving normal. *Simultaneity* is the separate question, and to read the relativity of synchrony as the denial of an objective simultaneity is a modal fallacy, the absence of a local test mistaken for the absence of a fact—the imperative's characteristic dual, established as such in the companion epistemic paper [7, 11]. This bears directly on the arguments that draw a metaphysical conclusion from the relativity of simultaneity. Rietdijk and Putnam argued that if the simultaneity of every inertial frame—its synchrony, in the terms used here—is equally valid, events on all of them are equally real, so that the future is as determinate as the past [77, 78]; Penrose's version sets two people passing in a street whose planes of simultaneity reach events in

the Andromeda galaxy days apart [80]. The premise is that no frame is physically distinguished, and Stein showed early that the relativistic structure alone does not license the conclusion [79]. What the data add is that in our universe the premise is false: the cosmic foliation is physically distinguished, by the content of the universe, and located by measurement. The lapse is the other part: the rate of advance of the layer that exists, what the clock measures [86]. To deny it—to hold the four-manifold itself the existent—is to treat occurrences as existents, which on analysis smuggles a fifth, meta-temporal dimension along which the block is supposed to "exist," and is incoherent as a basic description [86, 84]. The two corrections are not optional glosses on the formalism; they are what makes "a world that exists and evolves" a statable object at all. With them restored the measured result binds: the redshift isotropy forces the foliation whose advance the lapse is—a single cosmic time on whose slices the expansion is uniform—while the relativity of synchrony, untouched, is the shift, the existent the evolving layer the foliation orders and the manifold its record. What the isotropy does not by itself fix is how the rates of comoving clocks compare across regions today: across the last-scattering surface the measured anisotropy bounds their variation at the 10^{-5} level of the gravitational potentials, while at the observer's end any offset is common to every direction, so isotropy seen from one place does not settle it (§3.5). Accounts of an objective present built on cosmic time are developed by Ellis [32], by Ellis and Rothman [65] and by the contributors to [66]. Gödel objected that a cosmic time defined by the matter depends on how the matter happens to be arranged [67]; the measurement makes that dependence an empirical matter, settled for our universe.

Theorem 1 (The necessary and sufficient augmentation)**.** *Augmenting general relativity by fixing a physical foliation and reading it ontologically—the lapse the objective rate at which the existing world advances, the shift the relativity of synchrony—is both necessary and sufficient for a coherent formal description of an existing, evolving world.* Necessary: *the two parts are the pried-apart forms of the existence/occurrence and synchrony/simultaneity distinctions, and a description lacking the lapse part collapses existence into occurrence—the block reading, incoherent as a basic description.* Sufficient: *the two parts close the only coherent escapes—the "events exist" horn to the existence/occurrence correction as a category error, the "no objective present" horn to the synchrony/simultaneity correction as a modal fallacy, the latter then falsified outright by the measured isotropy—leaving the layered, cosmically-timed description with no surviving alternative. The augmentation alters none of general relativity's equations; it fixes which of the foliations they admit is physical and reads that one as the existent.*

The measurement is what turns this from a coherent option into a forced one: the lapse the theorem requires is the advance of a foliation that is not posited but measured, read directly off the redshift isotropy together with the uniformity of the expansion on its slices. The augmentation is thus not merely the structure a coherent description *may* take but the one the world is found to realize. What the theorem does *not* fix is which geometry realizes the augmented structure. It fixes that the structure is necessary and sufficient, and the measurement fixes that its necessary half obtains; which four-geometry carries it, and how matter sits on that geometry, are not settled here and are taken up in companions [4, 8, 9]. What is settled *here* is the structural verdict with its necessary half measured: the two-part augmentation—the completion of the Newton–Einstein formalism the standard reading left fused—is the necessary and sufficient one.

One qualification belongs with that scope, because the companion framework's cosmology sits inside it. The Friedmann models add a further posit the measurement does not by itself impose: that the slices of constant cosmic time are orthogonal to the matter's worldlines, so that cosmic time is the proper time of the matter. The redshift relation the floor constrains is also satisfied by a special non-synchronous case, in which every comoving worldline crosses a given slice of uniform expansion in the same way, the slices being tilted from the matter's rest frame; an example whose scale factor is identical to that of flat ΛCDM is given in [2]. On the metric

of that example the comoving congruence is sheared, and the observed isotropy is reproduced when the light is assigned to the worldlines at rest in the slices, by a reassignment of the metric's causal structure. Events on one slice are then simultaneous and yet asynchronous in the matter's rest frame: synchrony and simultaneity come apart for the matter itself, not only for observers moving through it. That case is the cosmology the companion framework constructs, where the reassignment is selected by the limiting structure of gravitational collapse rather than chosen, and where the direction along which each slice orders its comoving worldlines is a Killing field of the metric, the static one of the Schwarzschild–de Sitter family [4]. Nothing in the present argument depends on it. What the measurement forces is the foliation and a uniform expansion on its slices, and it forces them on either reading of how the matter's rest frame stands to the slices. Where the matter's worldlines are orthogonal to the slices, as the Friedmann models posit, the expansion of the matter is the mean curvature of the slices, so slices of uniform expansion are slices of constant mean curvature: the foliation the data select is then York's [68]. And how the matter's rest frame stands to the slices is what the source-count dipole tests (§1); a matter dipole exceeding the kinematic expectation would bear on that relation and not on the uniformity of the expansion the floor measures.

The warrant for the theorem's two moves is worth marking, because both are ordinary scientific inference rather than appeals to taste. Diagnosing the "events exist" horn as a category error is a correction of what the formalism actually asserts, not a preference about it. And preferring—as the floor above and the assessment below both do—the description that *requires* the observed structure over one that merely *permits* it is inference to the best explanation, the same step by which any theory that predicts a datum is preferred to one that accommodates it. Neither move is peculiar to this paper, and neither is taken on authority: each is made here on its own, and a companion paper takes up the general question of when such inferences are reliable [7]. *Nothing below draws on that treatment*, which is the point—an instance reached without the general account is a case that account can later rest on without circularity; and the category-error correction of §2 is the same method's first turn within this paper, so the paper's measured core and its structural verdict are operations of one grounded discipline.

It remains to weigh, against this result, the one objection that the uniform expansion is justified rather than assumed. It is held that Green and Wald have shown small-scale inhomogeneities to backreact negligibly on the large-scale metric, so that the Friedmann model is self-consistent with the averaged matter as its source [58, 21], against the contrary case that backreaction is dynamically significant [19, 20, 59]. Kept apart from the theorem and then set against it, the demonstration is seen to take as its input precisely what the floor forces: it assumes the metric already close to Friedmann–Lemaître–Robertson–Walker form—the cosmic time, the single global expansion—and proves the lumpiness does not disturb it. That is a statement about the *stability* of the assumed foliation, presupposing it and establishing its robustness, not deriving it—what is small in the demonstration is the metric perturbation ($\sim 10^{-5}$), while the matter contrast it rides on is nonlinear, so the licence to treat the configuration as a perturbed Friedmann model is itself a statement about the assumed background. The floor owes none of this, forcing the cosmic time and the uniformity of its advance from the redshift isotropy directly, a datum prior to any slice, with no Friedmann structure assumed in its derivation. This settles the standing dispute structurally and from outside it: the uniform expansion that one side assumes and the other questions is forced by measurement whichever way the averaging argument is adjudicated, so the most rigorous defence of the standard treatment is found to presuppose the very structure the measurement supplies, and the backreaction question, however it resolves dynamically, cannot reach the foliation the floor has fixed. Were the backreaction case to carry genuine large-scale content, it would enter on that merit and against this result; it does not enter as the ground of the uniform expansion, which the measurement has removed from its reach.

The scope is worth marking. This contradicts none of the standard model's fitted predictions;

with its assumptions in place it reproduces the expansion history observed, which is why it stands. What the measurement changes is the standing of those assumptions: the cosmic time and the uniformity of its advance are forced from the bottom up, prior to and independent of the dynamical fit, not conventions adopted for it. What the measurement does *not* do is install the replacement—the positive account, that the expansion is set by the cosmological constant alone with the matter density a reading of cosmic epoch [4], is motivated by this forcing but not compelled by it, and is left to the companion framework rather than asserted here.

This is the empirical counterpart, made quantitative, of the conceptual point that the cosmic present is measured rather than posited. As stellar parallax once converted the Earth's motion from an interpretive option into a measured fact, the isotropy of the cosmological redshift—with our own velocity through the rest frame read directly from the dipole, (369.82 ± 0.11) km/s toward Crater [48]—converts the cosmic foliation from a modelling convenience into a measured feature of the world. It excludes inhomogeneous cosmologies whose expansion rate tracks the local density, singles out the foliation that the 3+1 formulation treats as a free choice, and removes the physical premise of the arguments that infer from the relativity of simultaneity that no present is distinguished. The general result that no exterior temporal slicing can record a finite-time completion of gravitational collapse [6] stands on causal structure alone and needs none of this evidence; what the evidence adds is that the slicing that result concerns is not one option among many but the one the universe has let us measure. *Scope.* The temporal claim—uniform expansion throughout cosmic history—is clean, being the integral from decoupling to the present along the observed rays; the extension to a globally maximally symmetric space rides on the Copernican principle and is an extrapolation beyond the directly constrained region. *And the scope of what is claimed is worth stating against what is not.* What the isotropy measures is the uniformity of the expansion rate along the observed rays; it does not measure, and this paper does not claim, anything about which geometry carries that expansion or how the matter within it is arranged. The matter lumpiness is untouched. What is excluded is an inhomogeneous expansion *rate*, and the reason the two are separable is the reason the floor exists at all.

References

[1] W. de Sitter, *On the relativity of inertia. Remarks concerning Einstein's latest hypothesis*, Proc. Kon. Ned. Akad. Wetensch. **19** (1917) 1217–1225.

[2] D. Janzen, *A critical look at the standard cosmological picture*, in *Questioning the Foundations of Physics: Which of Our Fundamental Assumptions Are Wrong?*, A. Aguirre, B. Foster and Z. Merali eds., Springer, Cham (2015) 103–124, arXiv:1301.5980.

[3] D. Janzen, *Einstein's cosmological considerations*, arXiv:1402.3212.

[4] D. Janzen, *Collapsed matter must become a universe: the necessary and sufficient augmentation of general relativity into a description of structural existence, proven for gravitational collapse of any symmetry*, companion paper (P7).

[5] D. Janzen, *Reached through the imaginary, real by construction: the maximally symmetric substrate as the geometric–ontological core of Cosmological Relativity*, companion paper (P17).

[6] D. Janzen, *The event horizon is a metric singularity: a missing definition in general relativity, and the black-hole problems it dissolves*, companion paper (P1).

[7] D. Janzen, *The shadow of existence: scientific theory-choice as an empirically grounded discipline, calibrated on the historical record and on a live programme*, companion paper (P6).

[8] D. Janzen, *Covariance of geometries over the de Sitter substrate: the slicing operator, the vacuum kernel, and matter as the bend of the cut*, companion paper (P8).

[9] D. Janzen, *The range of the de Sitter slicing operator: surjectivity onto the symmetry-reducible sector of general relativity, the Kerr–NUT–(A)dS vacuum kernel, and the wall of inhomogeneity*, companion paper (P9).

[10] D. Janzen, *Beyond Space-Time*, (2025).

[11] D. Janzen, *Albert Einstein's Wild Goose Chase*, cosmiCave.org (2025).

[12] D. Janzen, *The Fortress of Misunderstanding Physics*, cosmiCave.org (2025).

[13] Planck Collaboration (N. Aghanim et al.), *Planck 2018 results. VI. Cosmological parameters*, Astron. Astrophys. **641** (2020) A6.

[14] Planck Collaboration (N. Aghanim et al.), *Planck 2018 results. V. CMB power spectra and likelihoods*, Astron. Astrophys. **641** (2020) A5; arXiv:1907.12875.

[15] A. Lewis, A. Challinor, and A. Lasenby, *Efficient computation of CMB anisotropies in closed FRW models*, Astrophys. J. **538** (2000) 473–476; arXiv:astro-ph/9911177.

[16] G. F. R. Ellis, *Relativistic cosmology*, in *Proceedings of the International School of Physics "Enrico Fermi", Course 47: General Relativity and Cosmology*, R. K. Sachs ed., Academic Press, New York (1971) 104–182; reprinted in Gen. Rel. Grav. **41** (2009) 581–660.

[17] G. F. R. Ellis, R. Maartens, and M. A. H. MacCallum, *Relativistic Cosmology*, Cambridge University Press, Cambridge (2012).

[18] S. Räsänen, *Light propagation in statistically homogeneous and isotropic dust universes*, J. Cosmol. Astropart. Phys. **02** (2009) 011; arXiv:0812.2872.

[19] T. Buchert, *On average properties of inhomogeneous fluids in general relativity: dust cosmologies*, Gen. Rel. Grav. **32** (2000) 105–125; arXiv:gr-qc/9906015.

[20] D. L. Wiltshire, *Cosmic clocks, cosmic variance and cosmic averages*, New J. Phys. **9** (2007) 377; arXiv:gr-qc/0702082.

[21] S. R. Green and R. M. Wald, *How well is our Universe described by an FLRW model?*, Class. Quantum Grav. **31** (2014) 234003; arXiv:1407.8084.

[22] E. Hubble, *A relation between distance and radial velocity among extra-galactic nebulae*, Proc. Natl. Acad. Sci. USA **15** (1929) 168–173.

[23] A. S. Eddington, *Space, Time, and Gravitation*, Cambridge University Press, Cambridge (1920).

[24] H. P. Robertson, *Relativistic cosmology*, Rev. Mod. Phys. **5** (1933) 62–90.

[25] H. P. Robertson, *Kinematics and world-structure*, Astrophys. J. **82** (1935) 284–301.

[26] A. G. Walker, *On Milne's theory of world-structure*, Proc. London Math. Soc. (2) **42** (1937) 90–127.

[27] J. D. North, *The Measure of the Universe: A History of Modern Cosmology*, Oxford University Press, Oxford (1965).

[28] A. Einstein, *Kosmologische Betrachtungen zur allgemeinen Relativitätstheorie*, Sitzungsber. Königl. Preuss. Akad. Wiss. (Berlin) (1917) 142–152.

[29] A. Friedmann, *Über die Krümmung des Raumes*, Z. Phys. **10** (1922) 377–386.

[30] G. Lemaître, *Un univers homogène de masse constante et de rayon croissant rendant compte de la vitesse radiale des nébuleuses extra-galactiques*, Ann. Soc. Sci. Bruxelles A **47** (1927) 49–59.

[31] R. K. Sachs and A. M. Wolfe, *Perturbations of a cosmological model and angular variations of the microwave background*, Astrophys. J. **147** (1967) 73–90.

[32] G. F. R. Ellis, *Physics in the real universe: time and spacetime*, Gen. Rel. Grav. **38** (2006) 1797–1824; arXiv:gr-qc/0605049.

[33] R. K. Barrett and C. A. Clarkson, *Undermining the cosmological principle: almost isotropic observations in inhomogeneous cosmologies*, Class. Quantum Grav. **17** (2000) 5047–5078; arXiv:astro-ph/9911235.

[34] C. Clarkson, B. Bassett, and T. H.-C. Lu, *A general test of the Copernican principle*, Phys. Rev. Lett. **101** (2008) 011301.

[35] D. J. Eisenstein et al., *Detection of the baryon acoustic peak in the large-scale correlation function of SDSS luminous red galaxies*, Astrophys. J. **633** (2005) 560–574; arXiv:astro-ph/0501171.

[36] A. G. Riess et al., *Observational evidence from supernovae for an accelerating universe and a cosmological constant*, Astron. J. **116** (1998) 1009–1038; arXiv:astro-ph/9805201.

[37] S. Perlmutter et al., *Measurements of* Ω *and* Λ *from 42 high-redshift supernovae*, Astrophys. J. **517** (1999) 565–586; arXiv:astro-ph/9812133.

[38] D. L. Wiltshire, *Exact solution to the averaging problem in cosmology*, Phys. Rev. Lett. **99** (2007) 251101.

[39] D. L. Wiltshire, *Average observational quantities in the timescape cosmology*, Phys. Rev. D **80** (2009) 123512; arXiv:0909.0749.

[40] M. A. Nazer and D. L. Wiltshire, *Cosmic microwave background anisotropies in the timescape cosmology*, Phys. Rev. D **91** (2015) 063519; arXiv:1410.3470.

[41] A. Seifert, Z. G. Lane, M. Galoppo, R. Ridden-Harper and D. L. Wiltshire, *Supernovae evidence for foundational change to cosmological models*, Mon. Not. R. Astron. Soc. **537** (2025) L55; arXiv:2412.15143.

[42] H. Weyl, *Zur allgemeinen Relativitätstheorie*, Phys. Z. **24** (1923) 230; English translation in Gen. Rel. Grav. **41** (2009) 1661.

[43] J. Ehlers, P. Geren, and R. K. Sachs, *Isotropic solutions of the Einstein–Liouville equations*, J. Math. Phys. **9** (1968) 1344.

[44] W. R. Stoeger, R. Maartens, and G. F. R. Ellis, *Proving almost-homogeneity of the universe: an almost Ehlers–Geren–Sachs theorem*, Astrophys. J. **443** (1995) 1.

[45] U. S. Nilsson, C. Uggla, J. Wainwright, and W. C. Lim, *An almost isotropic cosmic microwave temperature does not imply an almost isotropic universe*, Astrophys. J. Lett. **522** (1999) L1.

[46] P. J. E. Peebles and D. T. Wilkinson, *Comment on the anisotropy of the primeval fireball*, Phys. Rev. **174** (1968) 2168.

[47] G. F. Smoot, M. V. Gorenstein, and R. A. Muller, *Detection of anisotropy in the cosmic blackbody radiation*, Phys. Rev. Lett. **39** (1977) 898.

[48] Planck Collaboration (N. Aghanim et al.), *Planck 2018 results. I. Overview and the cosmological legacy of Planck*, Astron. Astrophys. **641** (2020) A1.

[49] Planck Collaboration (N. Aghanim et al.), *Planck 2013 results. XXVII. Doppler boosting of the CMB: Eppur si muove*, Astron. Astrophys. **571** (2014) A27.

[50] G. F. R. Ellis and J. E. Baldwin, *On the expected anisotropy of radio source counts*, Mon. Not. R. Astron. Soc. **206** (1984) 377.

[51] N. J. Secrest et al., *A test of the cosmological principle with quasars*, Astrophys. J. Lett. **908** (2021) L51.

[52] N. J. Secrest, S. von Hausegger, M. Rameez, R. Mohayaee, and S. Sarkar, *Colloquium: The cosmic dipole anomaly*, Rev. Mod. Phys. **97** (2025) 041001.

[53] T. Buchert, *Dark Energy from structure: a status report*, Gen. Rel. Grav. **40** (2008) 467.

[54] S. Räsänen, *Dark energy from back-reaction*, J. Cosmol. Astropart. Phys. **02** (2004) 003.

[55] E. W. Kolb, S. Matarrese, and A. Riotto, *On cosmic acceleration without dark energy*, New J. Phys. **8** (2006) 322.

[56] C. Clarkson, G. Ellis, J. Larena, and O. Umeh, *Does the growth of structure affect our dynamical models of the universe? The averaging, backreaction and fitting problems in cosmology*, Rep. Prog. Phys. **74** (2011) 112901.

[57] T. Buchert and S. Räsänen, *Backreaction in late-time cosmology*, Annu. Rev. Nucl. Part. Sci. **62** (2012) 57.

[58] S. R. Green and R. M. Wald, *A new framework for analyzing the effects of small scale inhomogeneities in cosmology*, Phys. Rev. D **83** (2011) 084020.

[59] T. Buchert et al., *Is there proof that backreaction of inhomogeneities is irrelevant in cosmology?*, Class. Quantum Grav. **32** (2015) 215021.

[60] M. J. Rees and D. W. Sciama, *Large-scale density inhomogeneities in the universe*, Nature **217** (1968) 511.

[61] M. I. Scrimgeour et al., *The WiggleZ Dark Energy Survey: the transition to large-scale cosmic homogeneity*, Mon. Not. R. Astron. Soc. **425** (2012) 116.

[62] R. Maartens, *Is the Universe homogeneous?*, Phil. Trans. R. Soc. A **369** (2011) 5115.

[63] P. Zhang and A. Stebbins, *Confirmation of the Copernican principle at Gpc radial scale and above from the kinetic Sunyaev–Zel'dovich effect power spectrum*, Phys. Rev. Lett. **107** (2011) 041301.

[64] A. Moss, J. P. Zibin, and D. Scott, *Precision cosmology defeats void models for acceleration*, Phys. Rev. D **83** (2011) 103515.

[65] G. F. R. Ellis and T. Rothman, *Time and spacetime: the crystallizing block universe*, Int. J. Theor. Phys. **49** (2010) 988.

[66] W. L. Craig and Q. Smith, eds., *Einstein, Relativity and Absolute Simultaneity*, Routledge, London (2007).

[67] K. Gödel, *A remark about the relationship between relativity theory and idealistic philosophy*, in *Albert Einstein: Philosopher-Scientist*, P. A. Schilpp ed., Library of Living Philosophers, Evanston (1949) 555.

[68] J. W. York, *Role of conformal three-geometry in the dynamics of gravitation*, Phys. Rev. Lett. **28** (1972) 1082.

[69] D. N. Limber, *The analysis of counts of the extragalactic nebulae in terms of a fluctuating density field*, Astrophys. J. **117** (1953) 134.

[70] R. Arnowitt, S. Deser and C. W. Misner, *The dynamics of general relativity*, in *Gravitation: An Introduction to Current Research*, L. Witten ed., Wiley, New York (1962) 227; arXiv:gr-qc/0405109.

[71] T. W. Baumgarte and S. L. Shapiro, *Numerical Relativity: Solving Einstein's Equations on the Computer*, Cambridge University Press, Cambridge (2010).

[72] K. V. Kuchař, *Time and interpretations of quantum gravity*, in *Proceedings of the 4th Canadian Conference on General Relativity and Relativistic Astrophysics*, G. Kunstatter, D. E. Vincent and J. G. Williams eds., World Scientific, Singapore (1992) 211.

[73] C. J. Isham, *Canonical quantum gravity and the problem of time*, in *Integrable Systems, Quantum Groups, and Quantum Field Theories*, L. A. Ibort and M. A. Rodríguez eds., Kluwer, Dordrecht (1993) 157; arXiv:gr-qc/9210011.

[74] H. Gomes, S. Gryb and T. Koslowski, *Einstein gravity as a 3D conformally invariant theory*, Class. Quantum Grav. **28** (2011) 045005; arXiv:1010.2481.

[75] P. Hořava, *Quantum gravity at a Lifshitz point*, Phys. Rev. D **79** (2009) 084008; arXiv:0901.3775.

[76] T. Jacobson and D. Mattingly, *Gravity with a dynamical preferred frame*, Phys. Rev. D **64** (2001) 024028; arXiv:gr-qc/0007031.

[77] C. W. Rietdijk, *A rigorous proof of determinism derived from the special theory of relativity*, Phil. Sci. **33** (1966) 341.

[78] H. Putnam, *Time and physical geometry*, J. Philos. **64** (1967) 240.

[79] H. Stein, *On Einstein–Minkowski space-time*, J. Philos. **65** (1968) 5.

[80] R. Penrose, *The Emperor's New Mind*, Oxford University Press, Oxford (1989).

[81] C. B. Collins and S. W. Hawking, *The rotation and distortion of the Universe*, Mon. Not. R. Astron. Soc. **162** (1973) 307.

[82] D. Saadeh, S. M. Feeney, A. Pontzen, H. V. Peiris and J. D. McEwen, *How isotropic is the Universe?*, Phys. Rev. Lett. **117** (2016) 131302; arXiv:1605.07178.

[83] A. J. Zhou, S. Dodelson and D. Scolnic, *The isotropy of cosmic expansion in the early and late Universe*, arXiv:2506.14878 (2025).

[84] D. Janzen, *The canonical problem of time as a category error: an empirically forced cosmic time, deparametrization, the dissolution of frozen dynamics, and a graviton quantization the de Sitter horizon closes without a free parameter—the seam where ħ enters gravity—in Cosmological Relativity*, companion paper (P10).

[85] D. Janzen, *Did Einstein Misconstrue Relativity's Real Meaning?*, cosmiCave.org (2025).

[86] D. Janzen, *Time Travel: A Five-Dimensional Trope*, cosmiCave.org (2025).

[87] D. Janzen, *Cosmogenesis: the Big Bang as a synthesis of Cosmological Relativity's deductively forced consequences, and the primordial light-element abundances it produces*, companion paper (P16).

[88] F. W. Bessel, *Bestimmung der Entfernung des 61sten Sterns des Schwans*, Astron. Nachr. **16** (1838) 65–96.

Appendix R Computational Receipts

Each computation marked R in the text is verified by a runnable receipt in the `receipts/` directory that ships with this work; status OK = verified (traced, run, output matches the claim). This appendix is generated from the receipt index, not maintained by hand.

P4R1 `P04_redshift_isotropy_floor` [OK]
sec:floor (redshift-isotropy floor: light crossing independent uniform cells of length R = 8 hˆ-1 Mpc, each expanding at its own rate H(1 - f delta/3), accumulates a redshift scatter of 1.6e-2 with the matter field's correlations along the ray (8.9e-3 for independent cells, the most accommodating case), some 390 (220) times the whole measured CMB anisotropy 4.1e-5, from the measured Planck 2018 TT spectrum -¿ matter-tracking expansion excluded). *Computes:* CAMB on the Planck 2018 base parameters (sigma_8 0.811, z_* 1089.9; growth rate read at k = 1e-3 h/Mpc); g = aHf/3; uncorrelated int dchi gˆ2 R sigma_Rˆ2 and correlated int dchi gˆ2 (1/2pi) int dk k P_R for R = 4..300; ˜1200 cells at R = 8; measured TT rms l = 2..2508; genuine-floor (correlation and growth weighting both raise it) + a density-independent rate gives zero *Bound:* linear separate-universe rate taken as the hypothesis; the comparator is the measured spectrum *Run:* `python3 receipts/P04_modern_parallax/P04_redshift_isotropy_floor.py`

P4R2 `P04_the_floor_exceeds_the_measured_temperature_spectrum_at_every_multipole_and_the_exclusion_move` [OK]
sec:floor, fig:floor (the floor multipole by multipole against Planck's MEASURED TT spectrum: the Limber angular power of a matter-tracking expansion exceeds the measured power at every multipole from l = 2 to 2500 by at least 2.35e4 (2e5 at the median) at R = 8 hˆ-1 Mpc, by at least 2.5e4 at R = 4 and 872 at R = 20; at R = 50 and 100 it still exceeds at every multipole (minima 20 and 1.3), and at R = 300 at every multipole up to l = 492, by more than 60 up to l = 100; the rms figures (8.9e-3 cells, 1.6e-2 correlated, R scan to 9.6e-4 and 24x) pinned against the measured rms 4.06e-5). *Computes:* CAMB on the Planck 2018 base parameters; the measured PR3 TT spectrum ('P04_floor_data/planck_TT_full.txt'); the 33 high-l multipoles with non-positive measured power left out of the minimum ratios; about 30 s *Bound:* Limber, inaccurate by tens of per cent at l ¡˜ 10, against margins of four orders *Run:* `python3 receipts/P04_modern_parallax/P04_the_floor_exceeds_the_measured_temperature_spectrum_at_every_mult`

P4R3 `P04_the_timescape_model_falls_inside_the_hypothesis_because_its_clock_postulate_does_not_keep_the` [OK]
sec:sw (the timescape model falls inside the hypothesis: its walls expand at two-thirds the rate of its voids in the volume-average time (Wiltshire 2009, eqs. 11, 14, 15, 38), and its clock postulate does not keep the difference out of the redshift, a photon gaining H_v or H_w per unit of volume-average time with the clock rates entering only at the path's ends; the mean reproduces the model's own eq. 38, and independent void/wall cells of 10, 30 and 50 hˆ-1 Mpc give an rms in ln(1+z) of 2.4e-2, 4.2e-2 and 5.4e-2, some 600 to 1300 times the whole measured anisotropy 4.1e-5 -¿ timescape excluded with the averaging programme). *Computes:* the tracker identities, Q two ways, the eq. 38 mean and the three rms figures asserted; the analytic variance against a 4000-ray Monte Carlo within 5 per cent; under 10s *Bound:* the most accommodating case, independent cells; correlations along the ray raise the scatter *Run:* `python3 receipts/P04_modern_parallax/P04_the_timescape_model_falls_inside_the_hypothesis_because_its_clock`

P4R4 `I20_the_redshift_is_a_conformal_killing_charge_on_the_null_cone` [OK]
sec:decomp (I20 — THE REDSHIFT IS A CONFORMAL-KILLING CHARGE ON THE NULL CONE. Integrable-systems field bake, probe I20. In conformal FLRW form d/deta is shown by explicit Lie derivative to be a CONFORMAL and not an exact Killing vector, so its charge is conserved only where p.p = 0; there it is a*omega, whose constancy IS 1+z = a_obs/a_em and whose logarithm is int H dt. Carries a MASSIVE CONTROL: the spatial momentum still falls as 1/a but a*E is not constant for m nonzero, and it recovers constancy in the massless limit – so the null-cone restriction is load-bearing and P04's line-of-sight integral is a statement about light.). *Computes:* runs clean; nine verdicts including a massive control that fails off the cone and recovers in the limit *Bound:* derived here; no parameter pinned *Run:* `python3 receipts/P04_modern_parallax/I20_the_redshift_is_a_conformal_killing_charge_on_the_null_cone.py`

P4R5 `R2_the_papers_correlation_figures_are_the_ones_nothing_checked` [OK]
sec:floor (R2 — P04's CORRELATION THEOREM. For N cells of average pairwise correlation

rhobar along the ray the variance of their sum is N sigma^2 (1 + (N-1) rhobar), exceeding the independent value for every rhobar ¿ 0, so the uncorrelated floor is a lower bound; checked symbolically and by simulation. CONTROL: a NEGATIVE rhobar must reduce the variance, or the positivity clause would be vacuous.). *Computes:* runs clean; four verdicts including the sign control *Bound:* derived here; the size of the rise on the measured spectrum is P04_redshift_isotropy_floor's *Run:* `python3 receipts/P04_modern_parallax/R2_the_papers_correlation_figures_are_the_ones_nothing_checke`

P4R6 `R50_the_long_wavelength_bullet_reverses_for_the_observable_and_the_corner_is_three_parts_per_mill` [OK]

sec:floor (P04's floor and its correlated value are variances along one ray, while the bounded quantity is the scatter ACROSS directions; every sightline is cast from the same observer, so a mode longer than the observable region is a monopole and cancels out of the anisotropy – the two agree for modes shorter than the path and part only for longer ones). *Computes:* casts sightlines from one origin through a plane-wave Gaussian field and measures BOTH the single-ray variance and the across-sky scatter: the ratio is 0.990 at kL in 100..1000 and 2.0e-4 at kL in 0.01..0.1, monotone between, stable under halving the modes, doubling the sightlines and changing the seed; then integrates a BBKS CDM spectrum and finds only 3.1e-6 of the single-ray variance below k = 1/L, with 95 percent of it above kL = 70 *Bound:* SUPPORTS P04 – the single-ray variance is the across-sky scatter wherever the variance lives. NOT CLAIMED: any value of the floor itself (P04_redshift_isotropy_floor owns that) *Run:* `python3 receipts/P04_modern_parallax/R50_the_long_wavelength_bullet_reverses_for_the_observable_and_the_co`

`B11_three_triples_two_identifications` [OK]

sec:cycloid (THREE three-fold structures and TWO open identifications carried in two registers that do not reference each other — the root triple is the shared vertex of both and appears in neither as such). *Computes:* asserts all three structures at source, P4s two-triples question and its {0,1,2} requirement, and that neither register cites the other (8 checks) *Bound:* NOT-A-PAPER-CLAIM — connects two open items; no identification established *Run:* `python3 receipts/L221_the_bridge/B11_three_triples_two_identifications.py`

`B12_the_two_triples_are_typed_differently` [OK]

sec:cycloid (the two triples are NOT one structure: the ROOT triple is the f=0 locus (roots 1/sqrt3, 1/sqrt3, -2/sqrt3, sum zero) while the CAUSAL triple is three VALUES of f each with one locus — three-element sets of different KINDS). *Computes:* asserts P4s question and its three-equally-spaced-values clause, computes the Nariai cubics roots and zero-sum and double root, matches PO-2s two distinguished roots, and counts the loci per level (8 checks) *Bound:* NOT-A-PAPER-CLAIM — a NEGATIVE result discharging the corpuss only DISCOVERABLE-PROOF *Run:* `python3 receipts/L221_the_bridge/B12_the_two_triples_are_typed_differently.py`

Appendix L The Knowledge Ledgers

Each claim marked ⌞ in the text rests on a field bake or the figure–theorem bake recorded in the named ledger, which ships with this work and carries the probes, the receipts, and the three registers kept apart — what bit, what bounced, and where the boundary is. A ledger is working apparatus, not a publication, and is cited here rather than in the bibliography. This appendix is generated from the ledger index, not maintained by hand.

L12 `HARMONIC_ANALYSIS_LEDGER.md` [field bake]
The harmonic-analysis field-bake ledger — what bit, what bounced, and the boundary. Second of the four fields ‘L-272‘’s re-survey left outstanding. ‘OWED‘ 622

L25 `STATISTICS_INFERENCE_LEDGER.md` [field bake]
The statistics/inference field-bake ledger — what bit, what bounced, and why. The last unbaked field in ‘THE_MATHEMATICS_REACH‘’s candidate set, and the one ‘A5.5‘ needs. ‘OWED‘ 622

L24 `PROBABILITY_LEDGER.md` [field bake]
The probability and stochastic-processes field-bake ledger — what bit, what bounced, and where the boundary is

L14 `INTEGRABLE_SYSTEMS_LEDGER.md` [field bake]
The integrable-systems field-bake ledger — what bit, what bounced, and where the boundary is